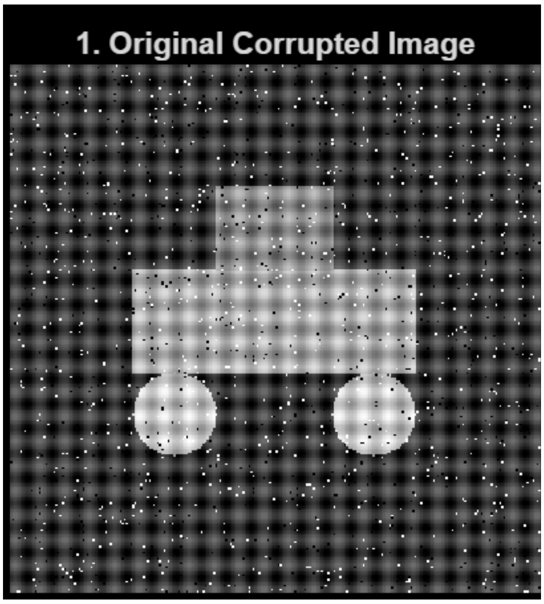
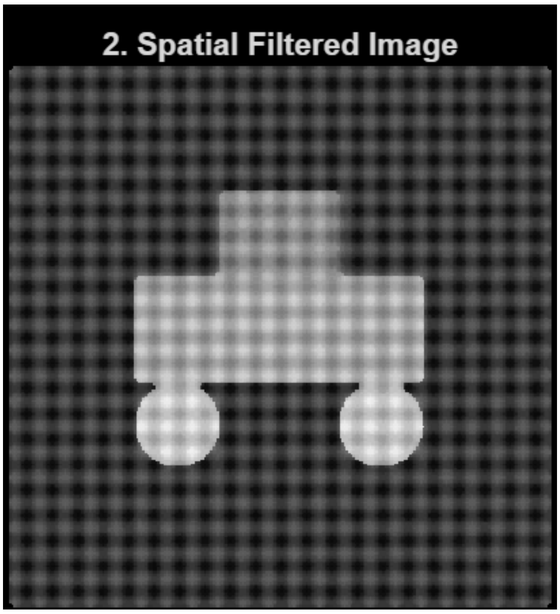
Signals Report



First we load the image, convert its values into float using *im2double* so that the transformations can be done accurately with marginal approximations and then we store it in a variable

1. Spatial Cleansing

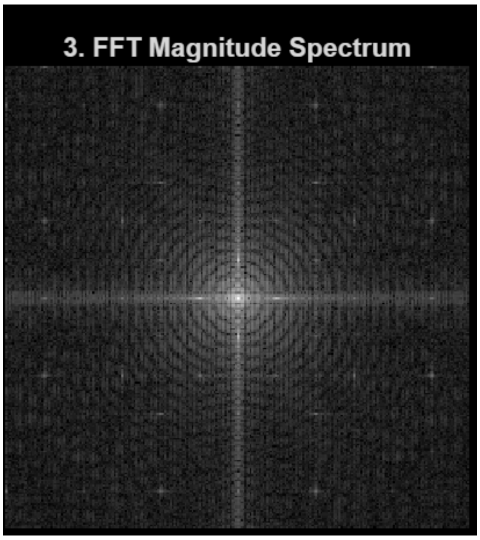
The image is rigged with salt-pepper noise. To resolve this we use Median Filter which is a standard technique to tackle such disruptions. *medfilt2* function is used to achieve this



2. Frequency Exorcism

To handle the gridlines visible in the image we apply 2D Fast-Fourier Transform which converts the image from spatial to frequency domain using *fft2* , *fftshift* to centralize the spectrum

abs calculates magnitude spectrum that helps with peak detection.
We can understand the points responsible for producing the grids easily as they appear in the form of bright spots



3. Automated Peak Detector

The central portion is part of the object so we apply a mask of radius 30 units to ensure the filter does not alter it
The center is identified in *cx*, *cy* then distance formula is applied

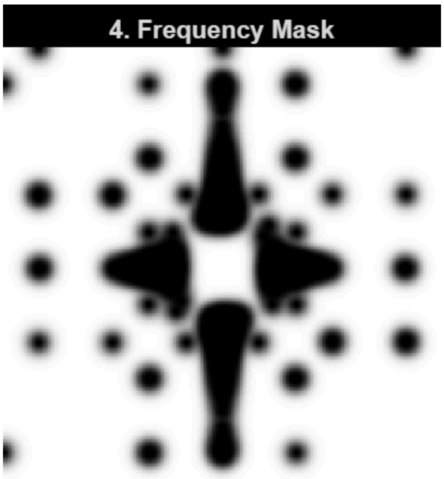
Using the concept of Z-Score Outlier Analysis and after some dynamic fine-tuning we observed *Mean+5*Std* to be an optimal threshold consistent across all 5 images. Any value exceeding this is flagged as an outlier and stored in *peak*

A Gaussian Notch filter with notch radius 5 is applied to shut out the detected interferences. Initially H is an all-pass filter consisting of only 1's. The loop iterates through every noisy pixel and the code smoothly attenuates it by altering the H value as:

$$H(u,v) = H(u,v) \cdot \left(1 - e^{-\frac{d^2}{2\sigma^2}}\right)$$

where sigma is the notch radius and d is the distance of a pixel from the noise

The final filter mask H looks like this:



This is then applied on the image in the frequency domain.

4. Restoration

The clean coordinates are reverted back to the spatial domain using Inverse FFT and the image is rescaled. A blueprint cmap is applied to produce the final processed output